Experiment -6

AIM:

To write a Program to implement error detection and correction using HAMMING code concept and make a test run to input data stream and verify error correction feature.

CODE:

```
def redundant = bits(m):
    r=0
    while 2**r< m+r+1:
        r+=1
    return r

def encode(data):
    m= len(data)
    r= rendundant_bits(m)
    Print ("number of data bits:",m)
    Print ("number of redandant bits:",r)

    n= m+r
    h= ['0'] *(n+1)
    j=0
    for i in range(r):
        p= 2**i
        Parity =0
        for j in range(1,n+1):
            if j& p:
                Parity ^= int (h [j])
        h[p] = str[Parity]
Print ("Reddundant bit Position:"[2**i for i in range(r)])
Print ("Hamming code:", "".join (h [1:]))
return "".join h [1:])
```

```
def check(h):
    n= len(h)
    r=0
    while 2**r < n
        r+ =1
    error =0
    Print("\n Parity Checking: ")
   for i in range (r):
        P=2**i
        parity=0
  for j in range (1, n+1):
       if j&P:
          Parity ^= int(h[j-1])
     Print("Position", P, ":", Parity)
     if Parity!=0:
         error += P
     if error ==0
         Print("\n no error detected")
     else:
         Print("Error has occured")
         Print("Error Position:", error)
       if h[error -1]=='0':
            h[error -1]="1"
       else:
            h[error -1]='0'
    Print("Error corrected")
    Print("corrected Hamming code:", "".join(h))
    return h
```

```
def remove_Parity (h):
        data = "
        for i in range (1, len(h)+1 ):
            if i & (i-1):
                data + = h[i-1]
            return data

def binary_to_text (data):
        text = ' '
        for i in range (0, len(data), 8):
            byte -data [i:i+8]
            text + = chr (int byte, 2))
        return text

text = input ("Enter data to transmit:")
binary = "join (format (ord(c), '086') for c in text)
Print ("\n Data:"text)
Print ("Binary:"binary)
channel = encode (binary)
with open ("channel.txt", "w") as f
        f.write (channel)
Print ("\n Data saved in chanel.txt")
with open (" chanel.txt ","r") as f:
        recived = f.read()
Print ("\n Recived data:")
Print (recived)
change = input ("\n Do you want to change the
        transmitted Data? (yes/no):")
if change.lower == "yes":
        Pos = int (input ("Enter Position to change:"))
        if Pos>=1 and Pos<=len (recived):
            h= list (recived)
```

```
if h[Pos-1]=='0":
        h[Pos-1]='1'
    else:
        h[Pos-1] = '0'
  recived = ''.join(h)
    Print("changed Data:")
    Print(recived)
  else:
    Print("Invalid Position")
    exit()
hamming =list(recived)
hamming = check(hamming)
data = remove_Parity(hamming)
Print("\n Data after removing redundant bits:")
Print(data)
output =binary_to_text(data)
Print("\n output:",output)
```

```
INPUT : Enter data to transit: HI
        Do you want to change the transmitted data?
        (yes/no) :no

output : Data : HI
         Binary: 0100100001001001
         no.of data bits :16
         no.of redundant bits:5
         Redundant bit Position : [1,2,4,8,16]
         Hamming code :1004 00110000 01 001001
         Data saved in Channel.txt
          Recived Data:
            100 1100110000001001001
          Data after removing redundant bits
           01001 0000
```

Result: Thus the Program is implemented using Python for error correction & error dection using hamming code was executed successfully.